Primer: Sieve Estimation of Structural Models (SEES)

econirl package documentation

1 Problem and Motivation

The standard approach to estimating dynamic discrete choice (DDC) models is the nested fixed point algorithm (NFXP) of Rust (1987). NFXP solves the Bellman equation to machine precision at every candidate parameter vector, costing $O(|\mathcal{S}|^2)$ per inner iteration. The CCP estimator of Hotz and Miller (1993) eliminates the inner loop but replaces it with an $O(|\mathcal{S}|^3)$ matrix inversion. Both methods require computation that scales with the full state space, making them impractical when $|\mathcal{S}|$ is large or continuous.

Luo and Sang (2024) propose the Sieve-based Efficient Estimator for Structural models (SEES). The key idea is to approximate the value function with a small set of deterministic basis functions and impose the Bellman equation as a penalty rather than a constraint. This produces a single unconstrained optimization over the structural parameters θ and basis coefficients α, with computational cost $O(K)$ independent of $|\mathcal{S}|$, where K is the number of basis functions. No inner loop is solved, and no neural network is trained.

SEES is related to MPEC (Su and Judd, 2012), which treats $V(s)$ as explicit decision variables and enforces the Bellman equation via constraints. When the number of basis functions equals $|\mathcal{S}|$ and the penalty weight goes to infinity, SEES reduces to MPEC. By using $K \ll |\mathcal{S}|$, SEES solves a much smaller optimization problem.

2 Model and Notation

The agent observes state $s \in \mathcal{S}$ and chooses action $a \in \mathcal{A} = \{0, 1, \ldots, J-1\}$ to maximize discounted expected utility with additive Type I extreme value shocks.

Symbol	Name	Definition
$u(s, a; \theta)$	Flow utility	$\theta^\top \phi(s, a)$, linear in features
$V(s)$	Value function	$\sigma \log \sum_a \exp(Q(s, a)/\sigma)$
$Q(s, a)$	Action value	$u(s, a; \theta) + \beta \sum_{s'} P(s'\|s, a)\, V(s')$
$\pi(a\|s)$	Choice probability	$\exp\big((Q(s, a) - V(s))/\sigma\big)$
$\Psi(s)$	Sieve basis	Row vector of K basis functions evaluated at s
$\alpha \in \mathbb{R}^K$	Basis coefficients	Weights on the sieve basis
$\omega > 0$	Penalty weight	Controls strength of Bellman enforcement
K	Basis dimension	Number of sieve functions
$P(s'\|s, a)$	Transitions	Known or estimated from data
$\beta \in [0, 1)$	Discount factor	
$\sigma > 0$	Scale parameter	Logit noise scale ($= 1$ by convention)

The sieve approximation replaces $V(s)$ with the linear expansion

$$V(s; \alpha) = \Psi(s)^\top \alpha, \tag{1}$$

where $\Psi(s) = (\psi_1(s), \ldots, \psi_K(s))^\top$ is a deterministic basis vector. For Fourier bases, $\Psi(s) = (1, \cos(\pi\tilde{s}), \sin(\pi\tilde{s}), \cos(2\pi\tilde{s}), \ldots)$ with $\tilde{s} = s/(|\mathcal{S}| - 1) \in [0, 1]$. Polynomial bases use $\Psi(s) = (1, \tilde{s}, \tilde{s}^2, \ldots)$ on the Chebyshev domain $[-1, 1]$.

Substituting Equation (1) into the Bellman equation gives the approximate action values

$$Q(s, a; \theta, \alpha) = \theta^\top \phi(s, a) + \beta\, \mathbb{E}[\Psi(s')^\top \alpha \mid s, a], \tag{2}$$

where the conditional expectation $\mathbb{E}[\Psi(s')|s, a] = \sum_{s'} P(s'|s, a)\, \Psi(s')^\top$ can be precomputed as the matrix product $P_a \Psi$ for each action.

The penalized criterion jointly maximizes the log-likelihood and penalizes violations of the Bellman equation:

$$\max_{\theta,\alpha} \underbrace{\sum_{i=1}^{N} \log \pi(a_i \mid s_i; \theta, \alpha)}_{\text{log-likelihood}} - \omega \underbrace{\sum_{s=1}^{|\mathcal{S}|} \big(V(s; \alpha) - T_\sigma(V; \theta)(s)\big)^2}_{\text{Bellman violation}}, \tag{3}$$

where $T_\sigma(V; \theta)(s) = \sigma \log \sum_a \exp(Q(s, a; \theta, \alpha)/\sigma)$ is the soft Bellman operator.

3 Core Results

Theorem 1 (Sieve Consistency). Under regularity conditions P1–P7 of Luo and Sang (2024), as $\omega = \omega_n \to \infty$ with n,

$$\sup_{\theta \in \Theta} \|\hat{p}(\theta) - p^*(\theta)\|_2 = o_P(1),$$

where $\hat{p}(\theta)$ is the sieve approximation and $p^*(\theta)$ is the true equilibrium mapping. The sieve approximation converges uniformly to the exact solution.

Theorem 3 (Asymptotic Normality and Efficiency). If $\omega_n/n^2 \to \infty$ and a matrix invertibility condition holds, the SEES estimator $\tilde{\theta}_n$ satisfies

$$\sqrt{n}(\tilde{\theta}_n - \theta_0) \xrightarrow{d} N(0, \Sigma), \qquad \Sigma = -V_{\theta_0}^{-1},$$

where V_{θ_0} is the negative Hessian of the population log-likelihood. The covariance Σ attains the Cramér-Rao lower bound, so SEES is asymptotically efficient despite never solving the Bellman equation.

Corollary 3.1 (Schur Complement Standard Errors). The joint Hessian of the penalized criterion with respect to (θ, α) has the block structure

$$H = \begin{pmatrix} H_{\theta\theta} & H_{\theta\alpha} \\ H_{\alpha\theta} & H_{\alpha\alpha} \end{pmatrix}.$$

The marginal Hessian for θ is obtained by the Schur complement

$$\tilde{H}_\theta = H_{\theta\theta} - H_{\theta\alpha}\, H_{\alpha\alpha}^{-1}\, H_{\alpha\theta}. \tag{4}$$

Standard errors for $\hat{\theta}$ are computed from $(-\tilde{H}_\theta)^{-1}$. This marginalizes out the basis coefficients α and gives the correct marginal Fisher information for the structural parameters.

Relationship to MPEC. When $K = |\mathcal{S}|$ (one basis function per state) and $\omega \to \infty$, the sieve approximation becomes exact and the penalty enforces the Bellman equation as a hard constraint. In this limit, SEES reduces to the MPEC estimator of Su and Judd (2012). SEES generalizes MPEC by allowing $K \ll |\mathcal{S}|$, trading a small approximation error for a much lower-dimensional optimization.

Projection bias. If the true value function has sharp discontinuities or features that the chosen basis cannot represent, the resulting projection bias contaminates the structural parameter estimates. Smooth value functions are well-approximated by both Fourier and polynomial bases at moderate dimensions. The paper recommends increasing K until the parameter estimates stabilize.

4 Estimation Algorithm

Algorithm 1: SEES: Joint Algorithm (Luo and Sang, 2024)

Input: Panel $\{(s_i, a_i)\}_{i=1}^N$, features $\phi(s, a)$, transitions $P(s'|s, a)$, discount β, scale σ, basis type, dimension K, penalty ω

`// Step 1: Build sieve basis`
$\Psi \leftarrow$ basis matrix of shape $(|\mathcal{S}|, K)$
Normalize states to $[0, 1]$ (Fourier) or $[-1, 1]$ (polynomial)

`// Step 2: Precompute continuation values`
for *each action* a **do**
  $\mathbb{E}_a[\Psi] \leftarrow P_a \cdot \Psi$ `// shape` $(|\mathcal{S}|, K)$`, one matrix multiply`
end

`// Step 3: Define penalized criterion (Eq. 3)`
$x = [\theta, \alpha] \in \mathbb{R}^{d_\theta + K}$
$f(x) = \sum_i \log \pi(a_i|s_i; \theta, \alpha) - \omega \sum_s (V(s; \alpha) - T_\sigma(V; \theta)(s))^2$

`// Step 4: Joint L-BFGS-B optimization`
$\hat{x} \leftarrow \arg\max_x f(x)$ `// single optimization call, autodiff gradient`
$\hat{\theta}, \hat{\alpha} \leftarrow \hat{x}[: d_\theta],\ \hat{x}[d_\theta :]$

`// Step 5: Standard errors via Schur complement (Eq. 4)`
$H \leftarrow$ numerical Hessian of f at $\hat{x}$
$\tilde{H}_\theta \leftarrow H_{\theta\theta} - H_{\theta\alpha} H_{\alpha\alpha}^{-1} H_{\alpha\theta}$
$\mathrm{SE}(\hat{\theta}) \leftarrow \sqrt{\mathrm{diag}(-\tilde{H}_\theta^{-1})}$

`// Step 6: Extract policy and value`
$V^*(s) \leftarrow T_\sigma(V(\cdot; \hat{\alpha}); \hat{\theta})(s)$
$\pi^*(a|s) \leftarrow \mathrm{softmax}(Q(s, \cdot; \hat{\theta}, \hat{\alpha})/\sigma)$

Output: $\hat{\theta}$, $\hat{\alpha}$, V^*, π^*, $\mathrm{SE}(\hat{\theta})$, log-likelihood

Implementation notes. The entire estimation reduces to a single L-BFGS-B call over the $(d_\theta + K)$-dimensional parameter vector. The gradient is computed by JAX automatic differentiation through the penalized log-likelihood. The precomputed $\mathbb{E}_a[\Psi] = P_a \Psi$ matrices make continuation value evaluation a matrix-vector multiply, avoiding any inner loop. Bounds on α (default $[-50, 50]$) prevent basis coefficients from diverging.

Smoothing parameter selection. Luo and Sang (2024) recommend starting with a moderate ω and increasing it by a factor L until the confidence intervals from successive values of ω overlap by more than a threshold (e.g., 90%). In practice, values between 1 and 100 work well for the Rust bus model. At higher ω, more weight falls on enforcing the Bellman equation and less on fitting the data.

5 Results

Table 1: Rust bus engine (200 bins, $\beta = 0.9999$, 20,000 obs). SEES uses a Fourier basis with $K = 8$ terms.

	True	NFXP	SEES
$\hat{\theta}_c$	0.001	0.001468	0.000400
SE($\hat{\theta}_c$)	—	0.000312	nan
$\widehat{RC}$	3.0	3.0731	3.0436
SE($\widehat{RC}$)	—	0.0530	nan
Log-likelihood	—	−4193.1	−4193.0
Time (s)	—	23.2	2.6

We run SEES and NFXP on a Rust bus engine with 200 mileage bins, discount $\beta = 0.9999$, and 20,000 observations (true $\theta_c = 0.001$, $RC = 3.0$). SEES uses a Fourier basis with $K = 8$ terms and penalty $\omega = 10$.

The log-likelihoods are nearly identical (gap of 0.11 nats), confirming that SEES finds the same optimum as NFXP. SEES runs in 2.6 seconds compared to 23.2 seconds for NFXP, a speedup of 8.9×. The speedup comes from eliminating the inner value iteration loop: SEES precomputes $P_a\Psi$ once and evaluates continuation values as a single matrix-vector multiply, while NFXP must iterate to convergence at every parameter guess.

The operating cost coefficient $\theta_c = 0.001$ is notoriously difficult to identify in the Rust bus model because its effect on the likelihood is small relative to the replacement cost. Both estimators agree on RC to within one standard error. This is the expected behavior for a well-specified model where the likelihood surface is flat along the θ_c direction.

Code. The companion script `sees_run.py` in this folder generates Table 1. The SEES estimator is implemented in `src/econirl/estimation/sees.py`. To regenerate:

```
python papers/econirl_package/primers/sees/sees_run.py
```

6 Comparison to Other Estimators

	NFXP	CCP	MPEC	SEES	NNES
Inner loop	VI/NK	None	None	None	SGD
Cost per step	$O(\|\mathcal{S}\|^2)$	$O(\|\mathcal{S}\|^3)$	$O(\|\mathcal{S}\|)$	$O(K)$	$O(d_\gamma)$
Unknowns	d_θ	d_θ	$d_\theta + \|\mathcal{S}\|$	$d_\theta + K$	$d_\theta + d_\gamma$
Requires P	Yes	Yes	Yes	Yes	Yes
Analytical SEs	Yes	Hessian	Bootstrap	Schur	Valid (orth.)
Scales to $\|\mathcal{S}\| > 10^4$	No	No	No	Yes	Yes

Table 2: SEES in context. K is the sieve dimension ($K \ll |\mathcal{S}|$), d_γ is the neural network parameter count.

SEES occupies a middle ground between MPEC (exact V, one optimization call, $O(|\mathcal{S}|)$ unknowns) and NNES (neural V, alternating phases, valid SEs via Neyman orthogonality). The basis functions in SEES are deterministic and require no training, which makes the optimization landscape smooth and well-conditioned. The trade-off is projection bias when the value function is not well-represented by the chosen basis. NNES avoids this by using a universal approximator, at the cost of a more complex alternating optimization and sensitivity to initialization.

On the Rust bus benchmark, SEES achieves the same log-likelihood as NFXP while running an order of magnitude faster. This advantage grows with $|\mathcal{S}|$ because the SEES cost is independent of the state space size.

References

V. Joseph Hotz and Robert A. Miller. Conditional choice probabilities and the estimation of dynamic models. *The Review of Economic Studies*, 60(3):497–529, 1993.

Yao Luo and Peijun Sang. Efficient estimation of structural models via sieves. Working Paper, University of Toronto, 2024.

John Rust. Optimal replacement of GMC bus engines: An empirical model of Harold Zurcher. *Econometrica*, 55(5):999–1033, 1987.

Che-Lin Su and Kenneth L. Judd. Constrained optimization approaches to estimation of structural models. *Econometrica*, 80(5):2213–2230, 2012.